Intellectual Property (IP) & Licensing

making MR methodology open source
(without getting in trouble)

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Disclaimer

no legal binding, no claims of correctness or guarantees of any kind …

- Based on my **personal** hands-on experience
  - Practical experience to share from 40+ patents
  - A number of open-source software packages
- Focus on practical aspects and **open science**
- Declare **no claim of correctness** or universal applicability

Intellectual Property, MR and Open Science

IP law is a tough topic…

Types of IP

- Patents, Trade secrets, copyrights…

Patents and open source

- What are patents
- Relevance to open-source MRI

Trade secrets

- Rules of conduct in the open world

Copyrights & Authorships

- Why open source licenses are important
- How to chose the right license

How do I deal with all that?

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Intellectual property (IP)

Some basic definitions…

IP is a category of property

- A variety of intangible creations of the human intellect
- Types of intellectual property vary by country

Best-known IP types:

- Patents
- Copyrights
- Trademarks
- Trade secrets

Relevant for **software**
(code sharing, open science)

Relevant for **inventions**
(science and technology)

An **invention** is a novel device, method, composition or process

- Either an <u>improvement</u> upon a machine, product, or process
  - E.g. for increasing efficiency or lowering cost
- …or an <u>entirely new concept</u>
- Some (<u>but not all</u>) inventions can be patented

Image: Science and Invention magazine cover, November 1928 (wikipedia.org)

What is a patent?

Basic definitions continued…

A **patent** is a type of intellectual property that gives its owner the legal right to exclude others from making, using, or selling an invention for a limited period of time in exchange for publishing an enabling disclosure of the invention*

- Patent is a mechanism to legally protect an invention
  - But there are other mechanisms…

The United States of America

The Commissioner of Patents and Trademarks

Has received an application for a patent for a new and useful invention. The title and description of the invention are enclosed. The requirements of law have been complied with, and it has been determined that a patent on the invention shall be granted under the law.

Therefore, this 5,860,492

United States Patent

Grants to the person(s) having title to this patent the right to exclude others from making, using, offering for sale, or selling the invention throughout the United States of America or importing the invention into the United States of America for the term set forth below, subject to the payment of maintenance fees as provided by law.

If this application was filed prior to June 8, 1995, the term of this patent is the longer of seventeen years from the date of grant of this patent or twenty years from the earliest effective U.S. filing date of the application, subject to any statutory extension.

If this application was filed on or after June 8, 1995, the term of this patent is twenty years from the U.S. filing date, subject to any statutory extension. If the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121 or 365(c), the term of the patent is twenty years from the date on which the earliest application was filed, subject to any statutory extension.

Acting Commissioner of Patents and Trademarks

Attest

*source: wikipedia.org

How patents are protected in practice?

Tangible actions on intangible goods…

In most countries, IP rights belong to private law

- No patent police
- No government-side monitoring

The patent holder must themselves detect infringement cases

- Ultimately they may need to sue the infringing party to enforce their rights

For occupational inventions the employer is the IPR owner

- Universities own the majority of researchers' patents

Patents are good for open science

Quite surprisingly so…

A patent **restricts** the others from practicing the invention

- But only for a **period of time**!
- Contains **precise** (and **reproducible**) description

Majority of national patent laws permit research use

- *Unfortunately the US research exception is one of the narrowest*

Eventually patents run out, from then on:

- These solutions stay in the **open domain**
- Well described, reproducible and **open to everyone**

The best way to guarantee that a technical solution stays in the **public domain** is to patent it and **give up the IP rights**

Patents vs. Trade Secrets

Nothing is secret about a patent once it is published…

In the German speaking circles there is a term "Patentgeheimins" which is often misinterpreted

- Only to keep it secret until the date of filing

There are no such thing as "patent secret"…

… but there are **trade secrets**

Trade (a.k.a. Company) Secrets

- Information that has either actual or potential economic value **by virtue of it not being known** generally
  - Famous examples: Coca-Cola recipe, WD40 oil mix,…
  - Generic examples: recipes, product designs, customer lists, pricing schedules, manufacturing techniques, technical data, source code, marketing strategies, sales techniques,…

- Real-life IP strategies of high-tech companies combine patents and trade secrets
  - Patents legally prohibit reproduction of the products
  - Trade secrets make it technically challenging

image: wikipedia.org

Trade Secrets & Reverse Engineering

- Reverse engineering is often legitimate (EU)
  - Research or personal interest
  - Interoperability or reproduction of functionality
  - No access to source code or technical data
  - *The law basis of reverse engineering differs significantly depending on jurisdiction and context*

- Exceptions / opposing examples
  - **Stolen** or **illegitimately** acquired product samples
  - **Explicitly forbidden** by a purchase or service contract
  - End-user license agreements (EULA), a non-disclosure agreements (NDA) or master research agreements (MRA) typically **prohibit reverse engineering**

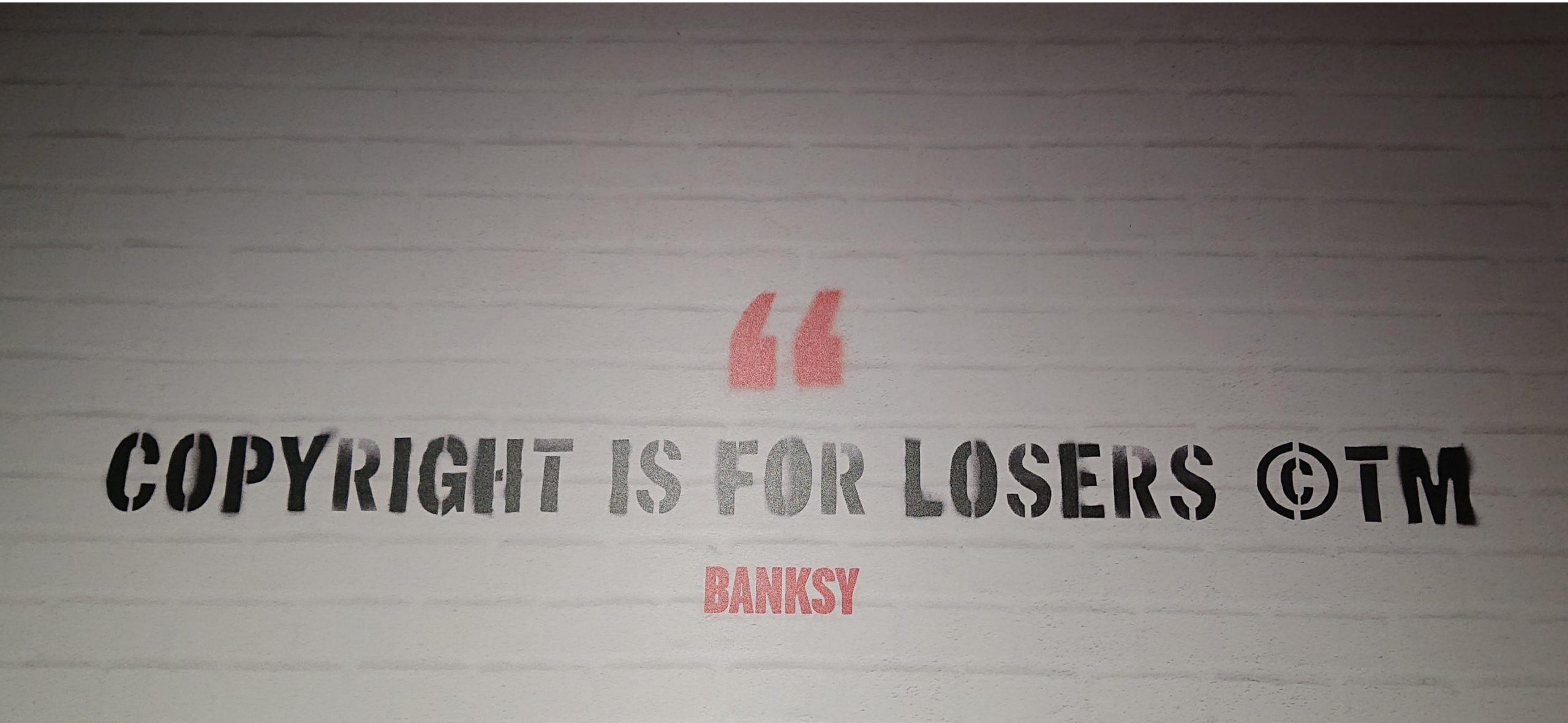
illustration by chatGPT

Trade Secrets & Scientific Collaboration

- “Slippery zone” for open science
- Non-disclosure contracts restrict disclosure
  - Implications for papers and open-source software
- Cooperation contracts
  - Commonly contain non-disclosure sections
  - Often explicitly forbid reverse engineering
- Medical technology research without industry?
  - Slow or even impossible translation to clinical applications

illustration by chatGPT

Copyrights?



Open Source Software Licenses Are Important!

- The art of giving!
  - Open-source software is not a gift
- Open source licenses are important!
- They are important for you
- They are important for users of your software
- They may constrain or promote dissemination

illustration by chatGPT

What Is Authorship?

- Everything that is published or distributed in public media **has an author**

- It is very difficult to stop being an author (legally)
  - Message in a bottle?
  - Doesn’t work with software…

- Authorship includes further things in the package
  - Copyright is **automatically** assigned to the author
  - Author bears some **responsibility** for their work

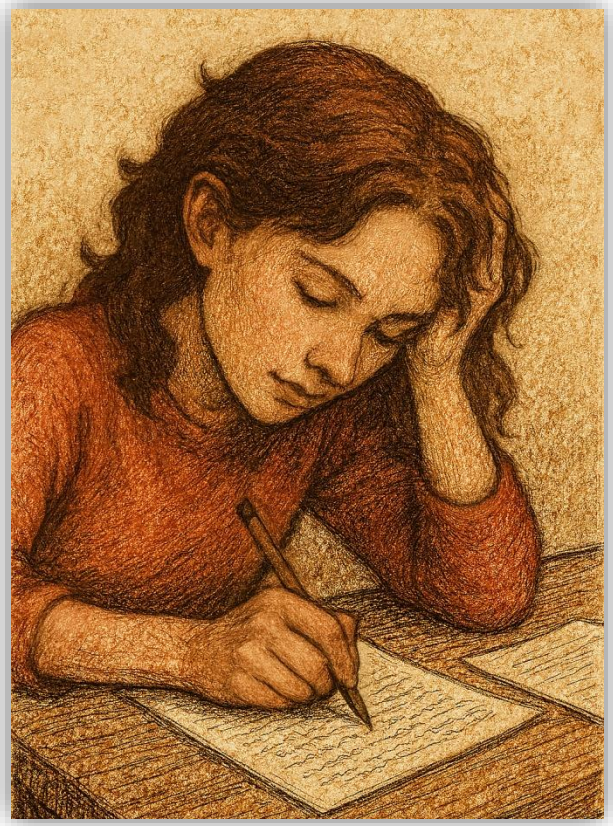


illustration by chatGPT

Complication of the Default Copyright

- Copyright belongs to the author
  - Even if not stated explicitly
- Copyright can be **enforced retrospectively**
  - By the author (e.g. if they change their mind)
  - By ancestors, relatives, employers, etc…
- Missing copyright release statement (a.k.a. copyright waiver) endangers both users and **authors of derived works**
  - Contradicts the open-science spirit

illustration by chatGPT

Author's Responsibility and Liability

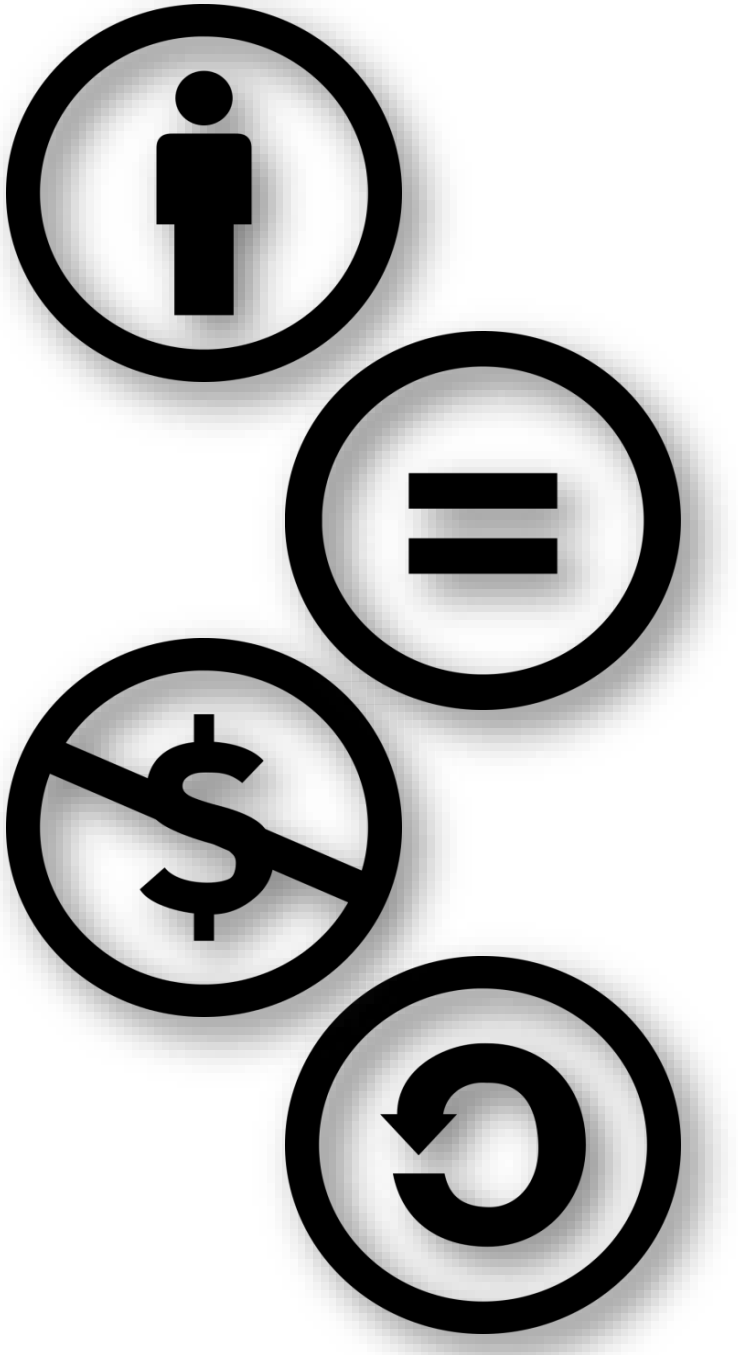
- In many countries consumer protection laws attribute **certain level of liability** also to "free" products, that has never been sold for money
- Authors of free software can potentially be sued for not fulfilling some expectations
- As an author of free software, you do not earn anything to compensate for the potential costs of the implied liability
  - Google "implied liability" to learn more

"Montagsauto" == "Lemon"
(German) (English)

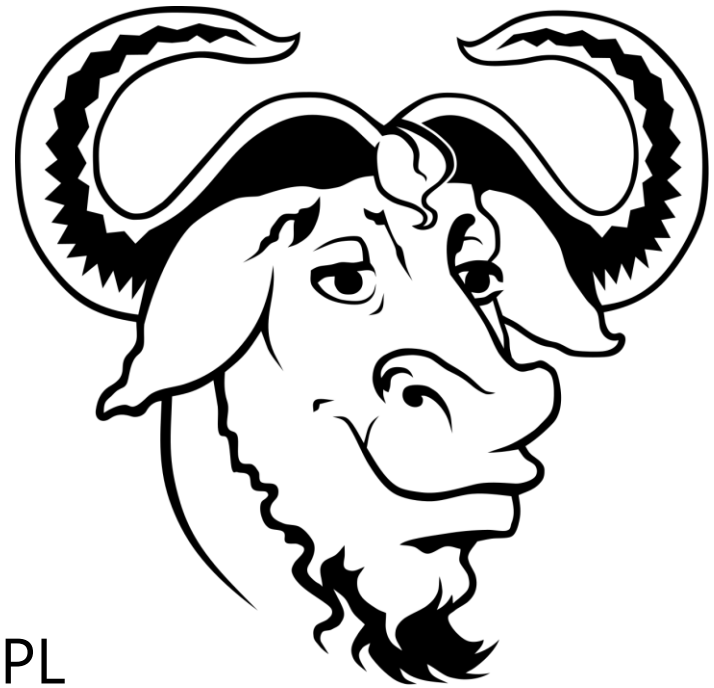
illustration by chatGPT

Open Source Licenses

- Open-source licenses regulate following aspects:
- Copyright release from the author(s)
- Disclaimer section
  - Protect the authors' from the implied liability
- Distribution, attribution and derived works
  - Permissive licenses: BSD, MIT, Apache; CC0 for non-software
  - Open non-commercial licenses: CC-BY-NC
  - *Copyleft (viral) licenses: GPL (GNU Public License), CC-BY-SA,…*



GNU General Public License (GPL)



- Played a key role in the establishment of the free and open-source software (FOSS)
  - Also known as a “copyleft” license
  - Authored by Richard Stallman in 1989
- A genius idea: what’s based on GPL software needs to be GPL
  - A **viral** spread of **enforced openness**
- Free Software Foundation: non-profit supporting free software movement
- Software industry has a great fear of GPL
  - GPL is known to **prevent collaboration** with device manufacturers
  - A big problem for biomedical research

- These days we recommend **permissive (MIT, BSD) or non-commercial** licenses

It's Time to Get Out from the Underground

- Open-source MRI is on the rise
  - Open source image reconstruction
  - Open-source MR pulse sequences (Pulseq & co)

Specialties of the MRI field

- Many patents
- Patent pool of major manufacturers
- Cooperation contracts
- Closed-source tradition

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Don’t Get Yourselves in Trouble

- Respect intellectual property
  - Know your rights and rights of others
- Understand and respect copyrights
  - Both own rights and rights of the others
- Educate yourself and your colleagues
  - Patents, product code vs. open science

illustration
by chatGPT

Handling patents of others

3rd party patents: handle with care!

You may infringe somebody's patent unknowingly

- Most likely case for researchers: open-source software

If you receive a letter from the IP owner…

- **Take it seriously**, talk to your technology transfer office
- Don't leave it unanswered!

Common misconceptions

- I have a patent on the same topic, which further develops the idea, so I don't have to license the original patent ➔NO
- Patents of 3rd parties can be practiced for any research ..?

Patents & Research Exemptions

Can I use any patent if I am a researcher?

The short answer is **No**!

- Some uses are permissible, but most aren't…

Permissible in most countries (except the US)

- Further R&D work to improve the original invention
- In the US, only research for pure curiosity is permissible; you are not allowed to extract any benefit from reproducing the invention

Not allowed: apply protected IP in other studies

- Apply patented medicine or procedures to patients
- Use patented technology to collect or analyze scientific data
- *Often **tolerated in reality** due to political, image or costs reasons*

If you publish an **open-source** implementation of protected technology costs of **lost revenue** can overweigh **court fees**

Patents and Open Source

- Open source implementation of protected IP can be problematic
- In theory, only the end users are the ones infringing on the patent
- In practice, both users and authors of the implementation can be sued
  - It is much easier to sue one author than the user community

- Handle with **respect**, but don't be afraid!
  - Check if your university has an **open-science policy**
  - If you aware of an active IP – **contact the inventor**
  - Also unintentional/unknowing IP infringement may result in claims
  - *Demand letter? Typically (!) resolved if the developer/researcher promptly* ***agrees to remove*** *the material from the public domain*

Carry the Message @IP_Owners

- Open-source implementations of the patented methodologies promotes them by **increasing exposure**
  - For methods that really work: independent validation
- Broad acceptance increases patent's **value**
  - Greater likelihood that vendors would license it
- 3rd party open implementations **save you work** while allowing for patent royalties from industry
- Universities & companies co-own inventions
  - Socially accepted rules of conduct
- We need a radical **change of culture** from protective **to permissive**

illustration by chatGPT

@Open_Science_Supporters: share responsibly

- **NEVER** copy from vendor's sources !!!
  - Publish YOUR code, not the code of the others

- Don't infringe on patents
  - Ask IP owners for a license
- Permission denied?
  Just state it in your paper!

DATA AVAILABILITY STATEMENT

MR pulse sequences for this publication were implemented in Pulseq and could potentially be made openly available. However, owners of the patent US 14,998,873 B2 have expressly forbidden us to share them freely on GitHub.

ey Online Library for rules of use; OA

- Don't be afraid
  - If you get a letter from the IP owner of which you didn't know before, just **remove your code** from the public repo

Open Source License Recommendations

- Creative Commons Zero (**CC0**) for text and artwork
  - Attribution (CC-BY) and non-commercial (CC-BY-NC) licenses are OK
- **MIT or BSD** are best for software
  - Compact, clear and permissive
- If you own IP (and don't want to give it away by the demo code)
  - Use the **3-clause BSD license**
- Even if your TTO insists on it, **avoid custom licenses**
- Use **no viral** licenses (GPL and alike) …

  if you want to be able to **work with the industry**

- **Medical technology is no-go without industry**

Thank you for your attention!

Time for questions?

Time, said Arthur weakly, is not currently one of my problems.
– Douglas Adams, The Hitchhiker’s Guide to the Galaxy